ODAY HAJ

Hardware Test Engineer | B.Sc. Electrical & Electronics Engineering
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SUMMARY

Hardware test engineer who defines, develops and automates tests for 4G/5G radio hardware at Parallel Wireless, and debugs failures to root cause across hardware, firmware and FPGA. In 15 months: delivered 50+ test-development stories, found 20+ hardware/firmware defects, and received the **"Feather in the Cap"** award. Uses AI tools (Claude Code) to move faster. B.Sc. in Electrical Engineering with VLSI and VHDL background. Looking to bring my test-development and hardware-debug experience to **silicon test**, improving coverage, test time and yield.

EXPERIENCE

Parallel Wireless *Jul 2025 – Present*

Validation Engineer – Hardware Test Development & Automation (RF Conformance)

★ **"Feather in the Cap" award** – company recognition for my contribution to hardware test automation.

- **Moved 7+ new hardware variants from lab to automated regression:** bring-up, calibration, test-plan review, automated test development, then daily CI/nightly runs. Tests follow 3GPP RF conformance specs (TS 36.141 / 38.141), run on Keysight signal generators and spectrum analyzers.
- **Increased test coverage:** wrote new tests for PA protection and PA shutdown, timing alignment across all bands, signal playback, new channel bandwidths and multi-radio configurations; added KPI pass/fail columns to reports and automatic device-log capture on every run.
- **Made the test system reliable and faster:** led the CI & Nightly Enhancements epic – zero aborted runs across all pipelines, stable results across all bands, and shorter run time.
- **Protected test integrity:** found and fixed false passes in receiver, calibration and firmware-upgrade test flows.
- **Root-caused 20+ hardware/firmware defects** with FPGA, firmware and PHY engineers – across receiver measurement accuracy, clocking and synchronization, transmit output power and data-converter behaviour.
- **Hands-on across the radio stack:** daily work on Linux, connecting to DUs and radio units over SSH, working directly with PHY and firmware to configure, test and debug.
- **Built a Python test-signal generator** that reverse-engineers vendor test waveforms and regenerates them for new bandwidths, bit-accurate to ±1 LSB, unblocking tests that had no vendor waveforms.
- **AI-accelerated engineering:** early adopter of **Claude Code** (AI coding agent) in daily test work – used it to reverse-engineer and regenerate test waveforms, write and review test code, analyze device logs, turn automation code into step-by-step lab procedures, and draft detailed Jira defect reports – cutting turnaround on new tests and investigations.
- **Supported multiple teams at once** (firmware, FPGA, PHY, DevOps, release): ran release regressions, answered test and setup questions, and set up Jenkins test pipelines with DevOps.
- **Ramped up new teammates** on test benches, tools and test flows, and wrote the manual test procedure for the lab.

EDUCATION

B.Sc. Electrical & Electronics Engineering – Braude College, Karmiel *2024*

Relevant: VLSI, VHDL, Electronics Labs 1 & 2, Signal Processing, Microcontrollers, Power Electronics. TA, Power Systems Lab.

Power Systems Completion Program – Kinneret College *2025*

ENGINEERING PROJECTS

- **VHDL Serial Communication System** – synchronous serial link with built-in error detection; simulated and verified in ModelSim, implemented in Quartus.
- **AI-Driven Drone (Final Project)** – real-time gesture/face-recognition control on Raspberry Pi 4; found and fixed the communication-lag bottleneck.
- **Low-Power IoT Sensor Device** and **Autonomous Cart** – reliable data over a lossy link; analog front end with current mirror and differential amplifier.

SKILLS

Test & Measurement: Keysight MXG / EXM / PXA, VNA, PathWave Signal Generation / Signal Studio, SCPI, Wireshark

Programming & Automation: Python (NumPy, pytest), Bash, VHDL, Jenkins, GitLab CI, Docker, Linux, SSH, Git, Jira

AI Tools: Claude Code (agentic AI coding assistant), prompt-driven automation and data analysis

Design Tools: Cadence, Quartus, ModelSim, LTspice, MATLAB, LabVIEW

Standards & Systems: 3GPP TS 36.141 / 38.141 (LTE / NR RF conformance), O-RAN fronthaul (NETCONF), PTP timing, DU / RU, PHY & firmware

Languages: Hebrew, Arabic, English